AWS | STUDENT PROJECT REPORT

VPC Peering

Connect two VPCs. Prove private connectivity.

Student name	**Aniket Kumar**
Batch	AWS Internship Batch 1
Project title	AWS VPC Peering (Project 01)
Region	Mumbai / ap-south-1
Completion date	15 / 10 / 2026
Email	aniketgaurav543@gmail.com

VPC A (ewb-vpc-a)	10.10.0.0/16	Public 10.10.1.0/24 (bastion) | Private 10.10.2.0/24 (App-A)
VPC B (ewb-vpc-b)	10.20.0.0/16	Private 10.20.1.0/24 (App-B) | no internet gateway
Peering	ewb-a-b-peer	pcx-requester (A requester, B accepter)

01 / ARCHITECTURE AND INVENTORY

Architecture, CIDRs and resource inventory

One internet entry point (the bastion in VPC A). Two private application servers, one per VPC, connected only through the peering connection. Administration path: laptop > bastion > App-A or App-B (SSH ProxyJump). Project traffic: App-A <> App-B over private IPs.

```
  Laptop --SSH/22--> [Internet Gateway ewb-igw-a]
                          |
+---------- VPC A 10.10.0.0/16 ----------+      +------ VPC B 10.20.0.0/16 ------+
| Public  10.10.1.0/24  ewb-bastion .10  |      |  (no internet gateway)          |
|        |  ProxyJump ------------------------------> App-B                       |
| Private 10.10.2.0/24  ewb-app-a   .10  <====> pcx <====>  10.20.1.0/24 ewb-app-b .10 |
+----------------------------------------+      +---------------------------------+
```

Why these CIDRs?

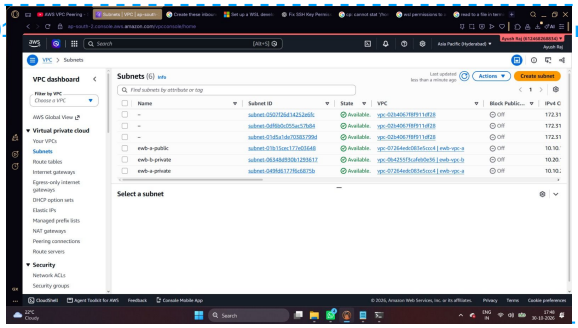
10.10.0.0/16 and 10.20.0.0/16 do not overlap, which is mandatory for peering. Each /24 subnet sits inside its own VPC range. Different subnet ranges alone would not fix overlapping VPC CIDRs.

Resource inventory (fill actual IDs)

Resource / Name tag	Location	CIDR / private IP	Actual ID
ewb-vpc-a	VPC A	10.10.0.0/16	vpc-
ewb-a-public	VPC A	10.10.1.0/24	subnet-
ewb-a-private	VPC A	10.10.2.0/24	subnet-
ewb-vpc-b	VPC B	10.20.0.0/16	vpc-
ewb-b-private	VPC B	10.20.1.0/24	subnet-
ewb-igw-a	attached to VPC A	-	igw-
ewb-rt-a-public / -a-private / -b-private	route tables	-	rtb- / rtb- / rtb-
ewb-sg-bastion / -app-a / -app-b	security groups	-	sg- / sg- / sg-
ewb-bastion	ewb-a-public	10.10.1.10 + public IPv4 _____	i-
ewb-app-a	ewb-a-private	10.10.2.10, no public IP	i-
ewb-app-b	ewb-b-private	10.20.1.10, no public IP	i-
ewb-a-b-peer	A to B	-	pcx-

All resources tagged Project = EWB-VPC-Peering and Student = Aniket Kumar. Key pair: ewb-lab (private key kept only on my laptop).

EVIDENCE: Checkpoint 01 — route table shows only local route



02 / SUBNETS AND ROUTE TABLES

Subnet CIDRs and the three route tables

ewb-rt-a-public / associated to ewb-a-public

Destination	Target	Purpose
10.10.0.0/16	local	Bastion to App-A
0.0.0.0/0	igw-ID (ewb-igw-a)	Bastion internet access
10.20.0.0/16	pcx-ID	Bastion SSH to App-B (ProxyJump)

ewb-rt-a-private / associated to ewb-a-private

Destination	Target	Purpose
10.10.0.0/16	local	Existing VPC A route
10.20.0.0/16	pcx-ID	App-A to VPC B

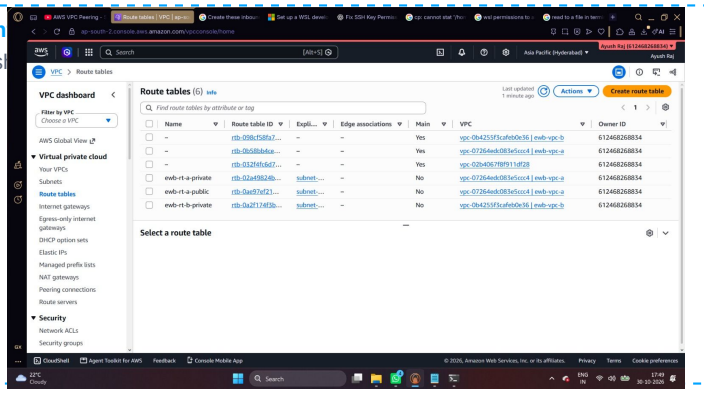
ewb-rt-b-private / associated to ewb-b-private

Destination	Target	Purpose
10.20.0.0/16	local	Existing VPC B route
10.10.0.0/16	pcx-ID	Return path to VPC A

Note: the private route table has no 0.0.0.0/0 route and VPC B has no internet gateway, so neither application server has internet egress. Peering is not transitive, so VPC B cannot use VPC A's internet gateway.

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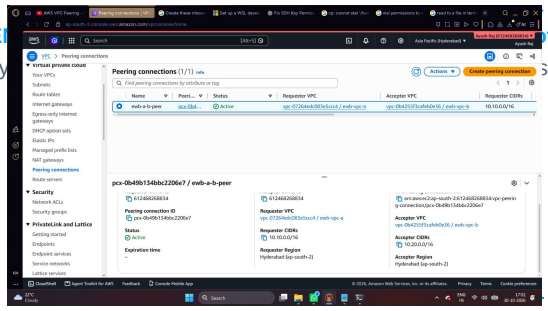
03 / PEERING AND SECURITY

Active peering and inbound security group rules

Field	Value
Name	ewb-a-b-peer
Requester VPC	ewb-vpc-a (10.10.0.0/16)
Accepter VPC	ewb-vpc-b (10.20.0.0/16)
Account / Region	My account / This Region (ap-south-1)
Status	Active (requested, then accepted)

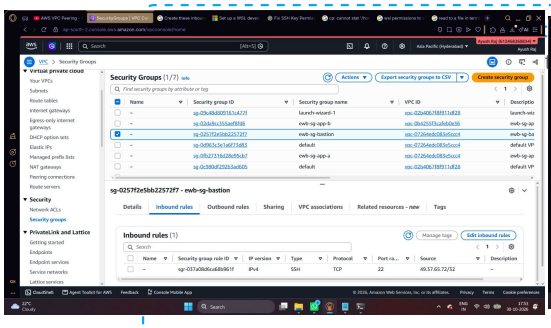
EVIDEN th VPC IDs

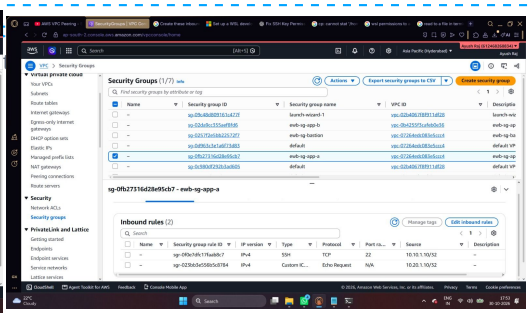
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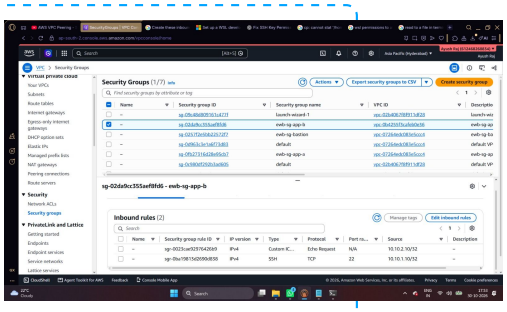


Inbound rules (outbound left at default allow-all; NACLs default allow-all)

Security group	Type / port	Source	Why
ewb-sg-bastion	SSH / TCP 22	My public IPv4 /32	Only I can reach the single public entry point
ewb-sg-app-a	SSH / TCP 22	10.10.1.10/32	Admin only via the bastion
ewb-sg-app-a	ICMP Echo request (type 8)	10.20.1.10/32	Allow ping from App-B only
ewb-sg-app-b	SSH / TCP 22	10.10.1.10/32	App-B sees the bastion's private IP as source
ewb-sg-app-b	ICMP Echo request (type 8)	10.10.2.10/32	Allow ping from App-A only







04 / VERIFICATION

Baseline, forward/reverse ping and private SSH

Local ~/.ssh/config used (all commands run on my laptop; key never copied to an instance):

```
Host ewb-bastion
    HostName <BASTION_PUBLIC_IP>
    User ec2-user
    IdentityFile ~/.ssh/ewb-lab.pem
    IdentitiesOnly yes

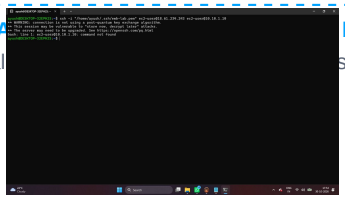
Host ewb-app-a
    HostName 10.10.2.10
    User ec2-user
    IdentityFile ~/.ssh/ewb-lab.pem
    IdentitiesOnly yes
    ProxyJump ewb-bastion

Host ewb-app-b
    HostName 10.20.1.10
    User ec2-user
    IdentityFile ~/.ssh/ewb-lab.pem
    IdentitiesOnly yes
    ProxyJump ewb-bastion
```

Test	Where / command	Expected (acceptance criteria)
01 Baseline	App-A, before peering: ping -c 4 10.20.1.10	No replies
02 Forward	App-A, after routes: ping -c 4 10.20.1.10	4 replies; 0% loss
03 Reverse	App-B, after routes: ping -c 4 10.10.2.10	4 replies; 0% loss
04 Private SSH	Laptop: ssh ewb-app-b	Shell on App-B via bastion
05 Isolation	EC2 console: App-A and App-B details	No public IPv4 on either

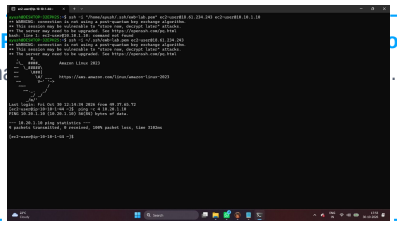
EVIDENCE: Test 01 - BASELINE FA **before peering (hostname, hostname -I, ping summary)**

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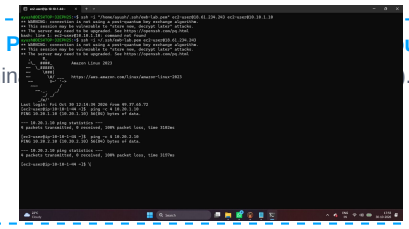
EVIDENCE: Test 02 - FORWARD P **utes**

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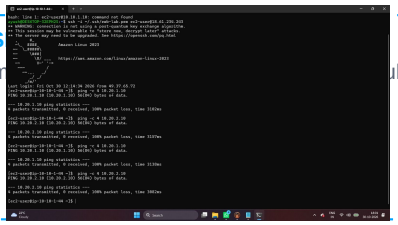
EVIDENCE: Test 03 - REVERSE P **utes**

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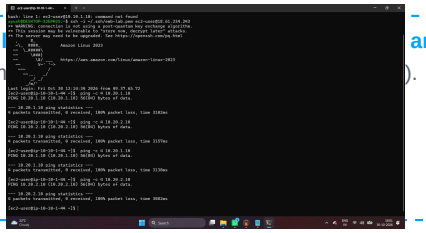
EVIDENCE: Test 04 - App-B SS

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EVIDENCE: Test 05 - EC2 detail **and App-B**

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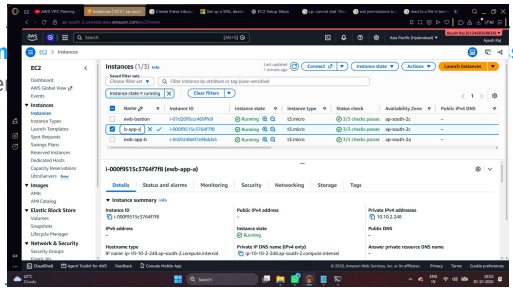
05 / CONTROLLED FAILURE

ICMP failure and recovery, with explanation

Procedure: remove only the App-B inbound ICMP Echo request rule (source 10.10.2.10/32), ping from App-A again (replies stop), restore the exact same rule, ping again (replies resume). The SSH rule stays in place throughout.

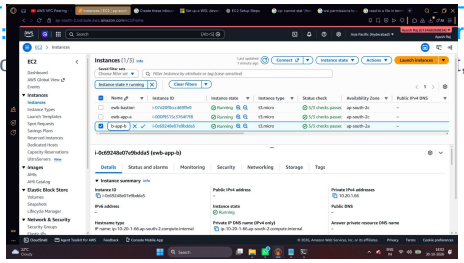
EVIDENCE: Failure: ping fron **S)**

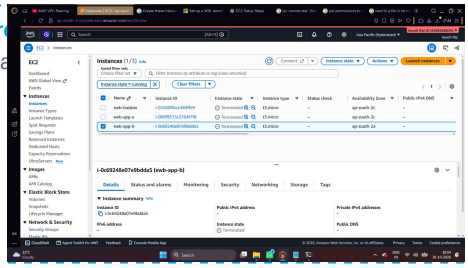
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Short explanation

The peering connection and both routes stayed Active/unchanged during the test, so the network path existed the whole time. Only the destination security group changed. Removing the Echo-request rule made App-B drop the packets, which proves that routes decide *where* traffic can go while security groups decide *whether* it is allowed. Restoring the exact rule brought replies back (security groups are stateful, so replies need no extra rule).

Forward path and return path

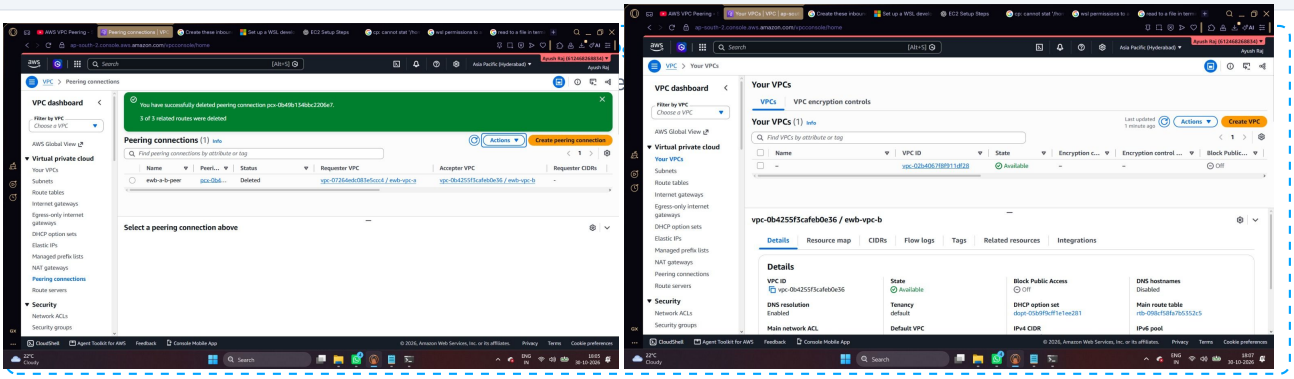
Forward (App-A to App-B): App-A (10.10.2.10) looks up 10.20.1.10 in ewb-rt-a-private, matches 10.20.0.0/16 and sends it to the peering connection; the packet reaches App-B, where ewb-sg-app-b allows Echo request from 10.10.2.10/32.

Return (App-B to App-A): App-B looks up 10.10.2.10 in ewb-rt-b-private, matches 10.10.0.0/16 to the same peering connection, and the stateful security group lets the reply out. Without the VPC B route the reply has no path back and ping fails.

07 / CLEANUP

Cleanup evidence and what I learned

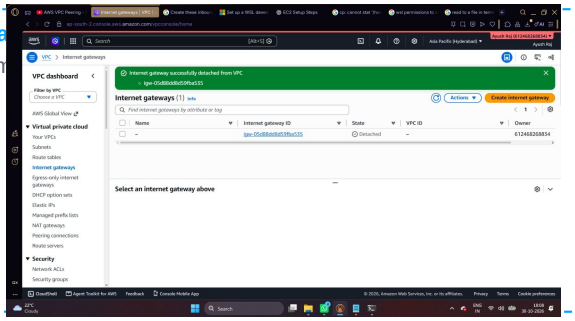
Step	Done?
Terminated ewb-bastion, ewb-app-a, ewb-app-b; EBS volumes deleted	[ ]
No Elastic IPs left allocated	[ ]
Peer routes and ewb-a-b-peer deleted	[ ]
Security groups, subnets, route tables deleted	[ ]
ewb-igw-a detached and deleted; both VPCs deleted	[ ]
Key pair ewb-lab and local .pem removed	[ ]
Filtered by tag Project = EWB-VPC-Peering: nothing remains; Billing checked	[ ]



What I learned

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TyIn this project I learned that VPC peering is only the link between two VPCs.
For traffic to flow, the peering connection must be Active, both VPCs need a route to the other VPC's CIDR, and the destination security group must allow the traffic.
Before peering, the ping from App-A to App-B failed.
After I added the routes and accepted the connection, it worked in both directions.
When I removed the ICMP rule from the App-B security group, the ping stopped even though the routes were fine,
so security groups can block traffic on their own.
I also learned that the bastion needs a route to VPC B,
because with ProxyJump the bastion opens the SSH connection to App-B itself.
Finally, I saw that peering does not give VPC B internet access,
and that deleting the resources afterwards avoids charges.